6 TIME AND WORK

6.1. Introduction

If a person can complete a work in ‘n’ days then he can do $\frac{1}{n}$ part of the work in one day.

The amount of work done be a person in 1 day is called his efficiency.

Example: A can do a work in 10 days. Then the efficiency of A is given by $A = \frac{1}{10}$.

Note: Number of days required to do a work = work to be done/work per day.

Example 1: If A can do a work in 10 days, B can do it in 20 days and C in 30 days in how many days will the three together do it?

Solution: The efficiencies are $A = \frac{1}{10}, B = \frac{1}{20}$ and $C = \frac{1}{30}$

So work done per day by the three = $\frac{1}{10} + \frac{1}{20} + \frac{1}{30} = \frac{11}{60} \Rightarrow No\ of\ days = \frac{60}{11} = 5.45\ days$.

Example 2: If A and B can do a work in 10 days, B and C can do it in 20 days and C and A can do it in 40 days in what time all the three can do it?

Solution:

$$A + B = \frac{1}{10}$$

$$B + C = \frac{1}{20}$$

$$C + A = \frac{1}{40}$$

Adding all the three we get $2(A+B+C) = \frac{7}{40} \Rightarrow A+B+C = \frac{7}{80} \Rightarrow No\ of\ days = \frac{80}{7}\ days$.

Note: If all the people do not work for all the time then the principle below can be used:

$$mA + nB + \boldsymbol{o}\boldsymbol{C} = \mathbf{1}. \quad \text{(1 is the total work)}$$

Here, m = no of days A worked

n = no of days B worked

o = no of days C worked

A, B, C = efficiencies

Example 3: If A can do a work in 12 days, B can do it in 18 days and C in 24 days. All the three started the work. A left after two days and C left three days before the completion of the work. How many days are required to complete the work?

Solution: Let the total no of days be x.

A worked only for 2 days, B worked for x days and C worked for $x - 3$ days.

So, mA + nB + oC = 1

$$\Rightarrow 2\left(\frac{1}{12}\right) + x\left(\frac{1}{18}\right) + (x-3)\left(\frac{1}{24}\right) = 1$$

$\Rightarrow$ 12 + 4x + 3(x-3) = 72

$$\Rightarrow x = \frac{69}{7}\ days.$$

Note: The ratio of dividing wages = ratio of efficiencies = ratio of parts of work done

Example 4: A can do a work in 10 days and B can do it in 30 days and C in 60 days. If the total wages for the work is Rs. 1800 what is the share of A?

Solution: Ratio of wages = $\frac{1}{10} : \frac{1}{30} : \frac{1}{60} = 6 : 2 : 1$ (Multiplying each term by LCM 60)

So total 9 equal parts in Rs. 1800 $\Rightarrow$ each part = Rs. 200 $\Rightarrow$ share of A = 6 parts = Rs. 1200.

Note: When pipes are used filling the tank they are treated similar to the men working but some outlet pipes emptying the tank are present whose work will be considered negative.

Example 5: A pipe can fill a tank in 5 hrs but because of a leak a the bottom it takes 1 hr extra. In what time can the leak alone empty the tank?

Solution: Let the filling pipe be A.

$$A = \frac{1}{5}$$

But with the leak L, $A - L = \frac{1}{6}$ (A-L because leak is outlet)

So, $\frac{1}{L} = \frac{1}{5} - \frac{1}{6} = \frac{1}{30} \Rightarrow$ Leak can empty the tank in 30 hrs.

❑❑❑